PG No. 40 $C_{4\mathrm{h}}(c)$ $4/m$ [tetragonal]

Table 1: Representation matrices

Irrep.						
A_g	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}1\end{bmatrix}$				
B_g	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$				
$E_g^{(a)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}-i\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}i\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}-i\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}i\end{bmatrix}$				
$E_g^{(b)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}i\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}-i\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}i\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}-i\end{bmatrix}$				
A_u	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$				
B_u	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}1\end{bmatrix}$				
$E_u^{(a)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}-i\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}i\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}i\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}-i\end{bmatrix}$				
$E_u^{(b)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}i\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}-i\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}-i\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}i\end{bmatrix}$				